

Module 1 – Inbound Security

Module 1 – Inbound Protection

Overview

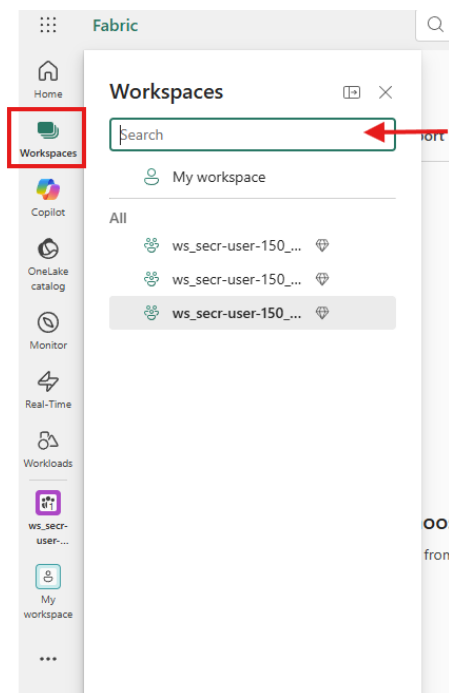
In this module, you will use pre-provisioned workspace **ws_<username>_wspl** to configure workspace private link (WS PL). Once configured, you will then test inbound access.

Exercise 1: Configure and deploy Workspace Private Link (WS PL)

In this exercise, you will be using Microsoft Fabric and Azure Portal to configure Fabric workspace private link (WSPL).

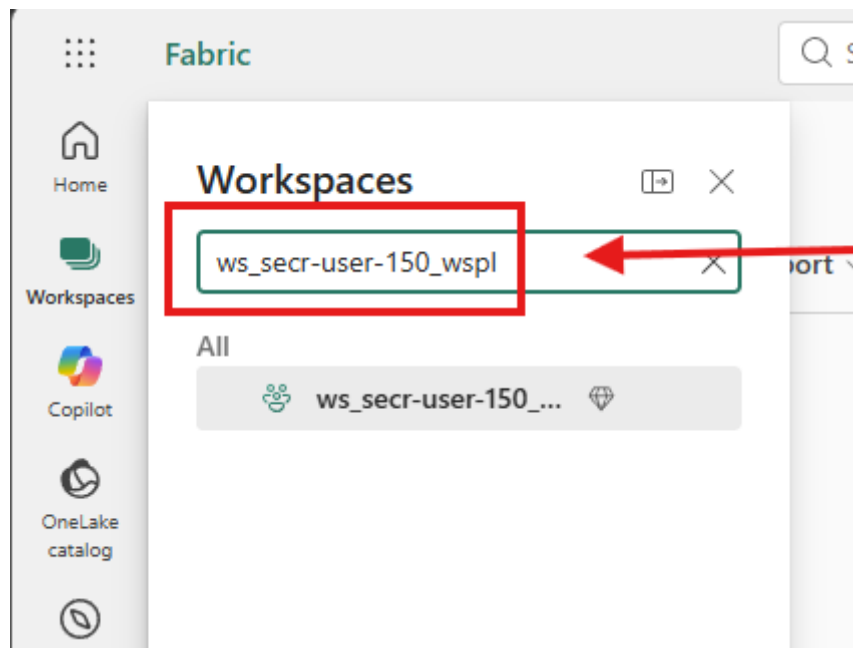
Step 1.1: Gather Details required to configure Workspace Private Link

1. If you are already inside Microsoft Fabric signed with your username for this workshop, go to next bullet. If not, open a new InPrivate (or InCognito) window and Login to Microsoft Fabric via <https://app.fabric.microsoft.com/>
Username: secr-username@fabcon26.onmicrosoft.com
(example: secr-user-150@fabcon26.onmicrosoft.com)
Password: Your password (that you setup in the previous step: “Before you start”)
2. Go to **Workspaces** in the Explorer pane on the left.

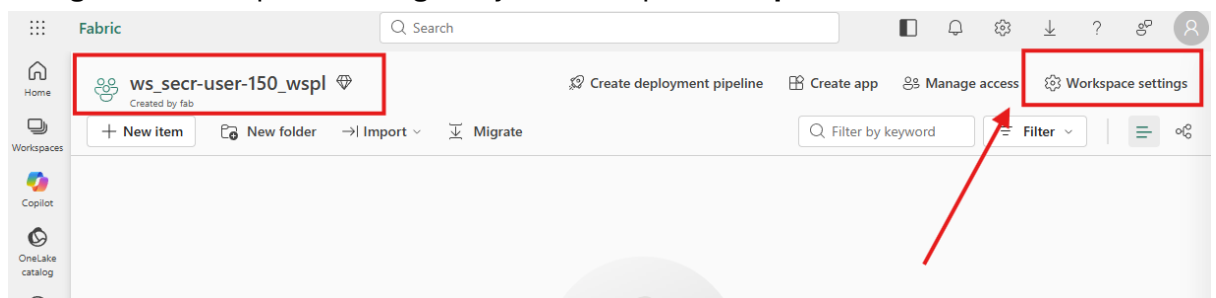


3. Search for your workspace. Name of the workspace assigned to you follows the naming pattern **ws_<username>_wspl**. Please ensure that you use workspace assigned to you with suffix **_wspl**.

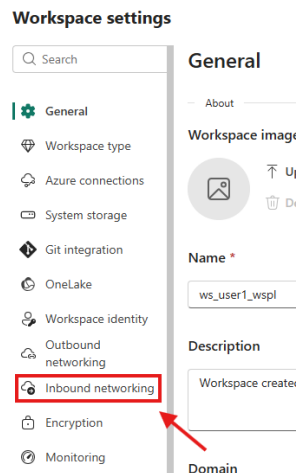
As shown here, if user account assigned to you is **secr-user-150** then workspace name will be **ws_sec-user-150_wspl**.



4. The workspace should contain pre-provisioned lakehouses along with sample data.
5. Navigate to Workspace Settings for your workspace **_wspl**



6. Select **Inbound Networking**.



7. Leave **Allow all connection to this workspace (including private links)**.

Note: Your instructor will discuss the two settings available under Inbound Networking. The Apply button will be greyed out because no change has been made. By default, where tenant-level PL and public access is not blocked at tenant-level, each workspace have public IPs, and traffic can be routed to the public endpoint. While keeping public endpoint routable, you can also enable Workspace PL in parallel and access workspace via private endpoint.

The second setting blocks public access to the workspace and restricts inbound access **ONLY** via private endpoint. For this workshop, since we do not have access to a jump-host, we will not block public access, and we can still demonstrate result of deploying WS PL without blocking public access.

Inbound networking

Apply workspace settings to manage inbound connections, deployment permissions, and Git access.

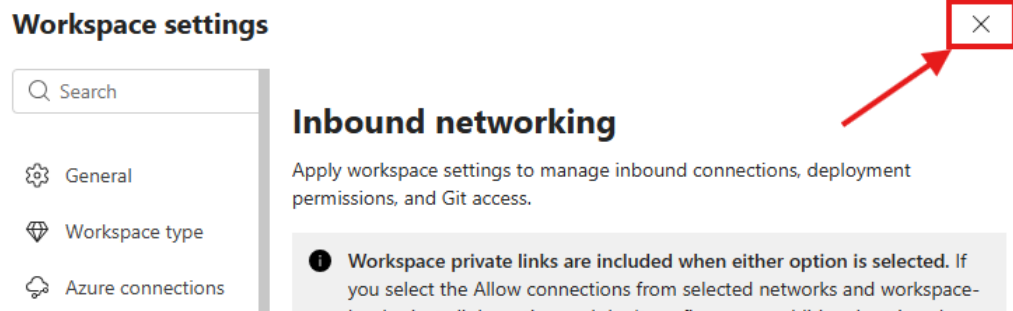
Workspace private links are included when either option is selected. If you select the Allow connections from selected networks and workspace-level private links option and don't configure any additional settings here, this workspace can only be accessed using workspace-level private links. To use workspace private links, you need to complete additional set up steps in Azure. [Learn more](#)

Workspace connection settings

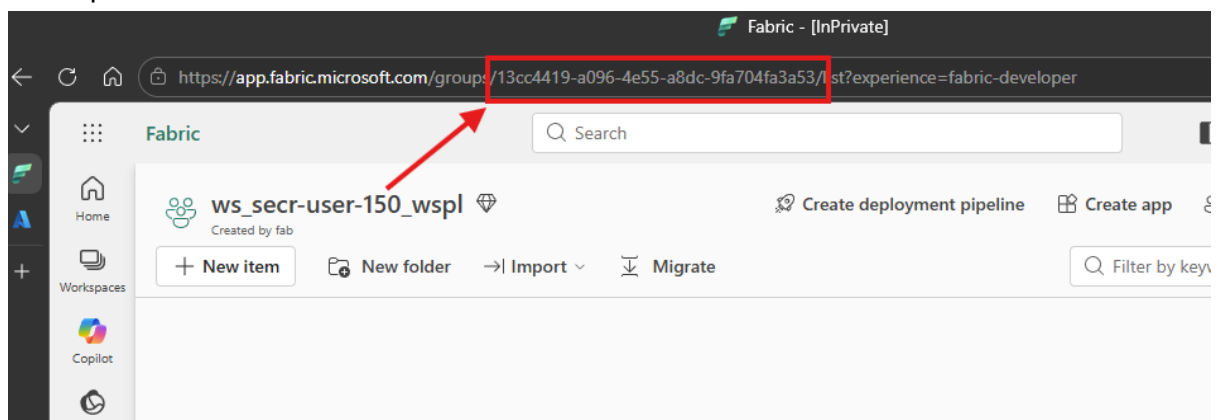
Choose how users and services can connect to this workspace. Use these settings to manage access, security, and integration options. [Learn more](#)

- ☒ **Allow all connections to this workspace (including private links)** ⓘ
Users can connect from any network, including through private links. This setting may be limited by configurations set by your Fabric admin.
- ☐ **Allow connections from selected networks and workspace level private links**
Access to this workspace is restricted to only the IP addresses specified below.

8. Without making any changes to the settings, exit from **Workspace Settings** by clicking **X** button at the top.

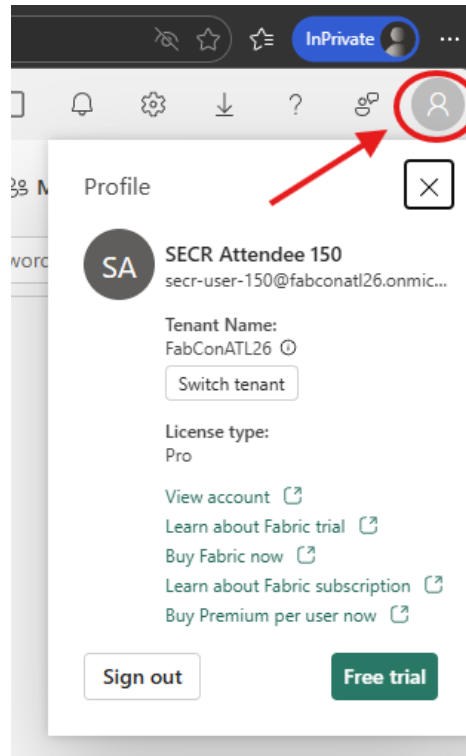


9. Note down the workspace ID of your workspace **_wspl**. One way to find out the workspace ID is to use the address in the browser web address bar.

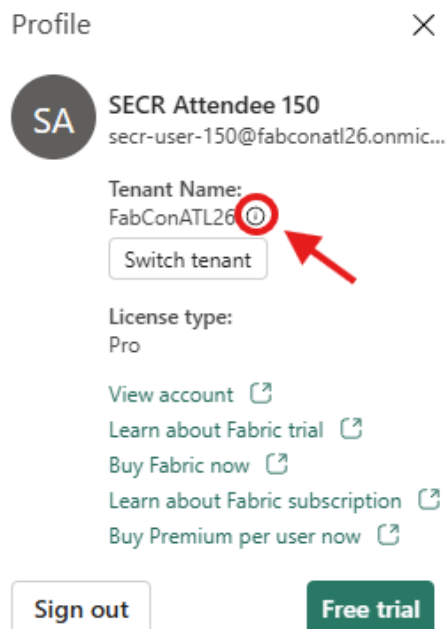


In this example, workspace ID is **13cc4419-a096-4e55-a8dc-9fa704fa3a53**. **In your case, it will be different.** You will need workspace ID to configure workspace private link in Azure. Make a note of it in notepad.

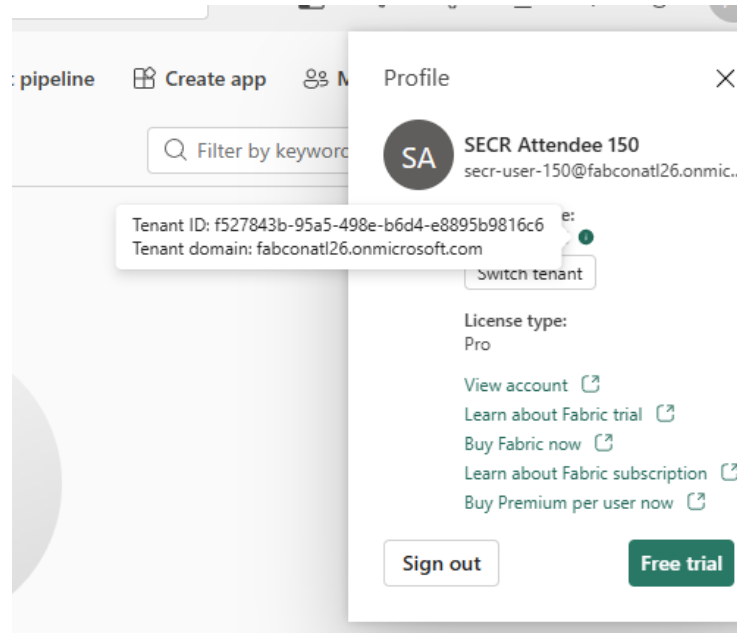
10. To find out **Tenant ID** of your deployment, click on your user profile at the top-right corner of Fabric web portal.



11. Hover the mouse cursor Information, and the pop-up should appear with the tenant id details of your Fabric tenant.



12. The tenant ID is listed under tenant name. You can copy and paste it to a notepad as you'll be using it later to configure workspace PL in Azure.



Tenant ID: **f527843b-95a5-498e-b6d4-e8895b9816c6**

The Tenant ID is common for all users in this workshop.

Checkpoint – At this stage, you have a workspace, workspace ID, Tenant ID and cross-checked Workspace Settings ensuring that Public Access is not blocked to the workspace.

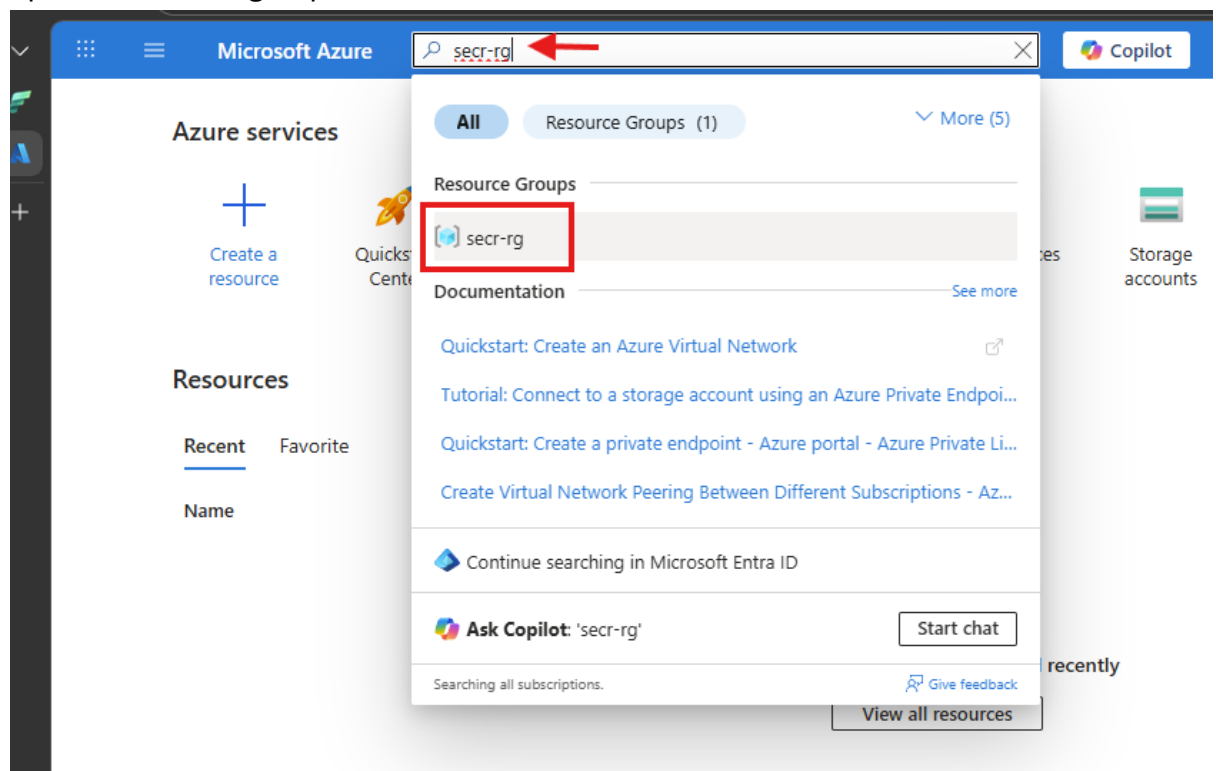
Step 1.2: Provision Private Link Service (PLS) in Azure for your Workspace

In this section, you will deploy resources on Azure to enable workspace PL and provision associated endpoint for your workspace.

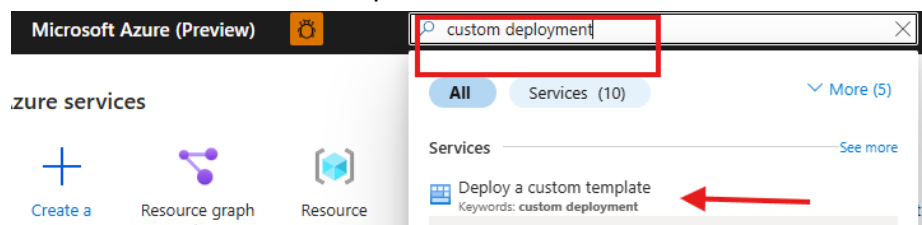
13. If not already opened, please login to Azure Portal via

<https://www.portal.azure.com>

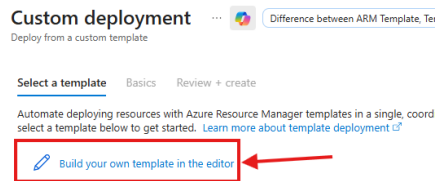
14. In the Azure Portal, navigate to resource group (RG) pre-provisioned for this exercise. Search for **secr-rg** (search for secr-rg in the search bar at the top) and open the resource group.



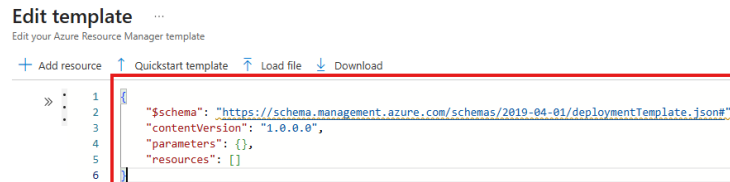
15. From the Azure portal search bar, search for **deploy a custom template** and then select it from the available options.



16. On the **Custom deployment** page, select **Build your own template in the editor**:



17. In the **Edit Template** experience, you can select and remove pre-filled template.



18. Copy the following code snippet and paste it in template and **replace your resource-name, workspace-id and tenant-id**
(Note – replace your <tenant-id> and <workspace-id> with actual values that you obtained from the previous steps). For resource-name use the following format: **pls-ws-<username>-wspl (only hyphens are permitted, no underscores)**

For example, if your username is SECR-USER-150, then use pls-ws-secr-user-150-wspl

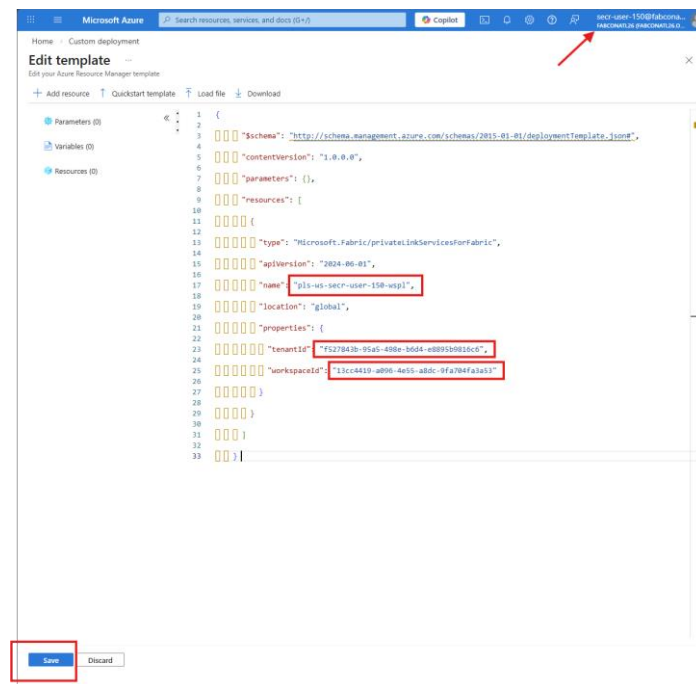
<workspace-id>: <your workspace ID from previous step>

<tenant-id>: f527843b-95a5-498e-b6d4-e8895b9816c6 (common for all)

<resource-name>: pls-ws-<username>-wspl

```
{
  "$schema": "http://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {},
  "resources": [
    {
      "type": "Microsoft.Fabric/privateLinkServicesForFabric",
      "apiVersion": "2024-06-01",
      "name": "<resource-name>",
      "location": "global",
      "properties": {
        "tenantId": "<tenant-id>",
        "workspaceId": "<workspace-id>"
      }
    }
  ]
}
```

After replacing the values, your template will look like this. Once done making changes, select **Save**.



19. In the **Resource Group** dropdown, select your resource group name. Please use the pre-provisioned resource group assigned to you – **secr-rg**

Select a template **Basics** Review + create

Template

 Customized template [↗](#)
1 resource

[Edit template](#)  [Visualize](#) 

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

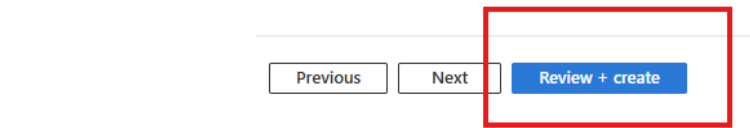
Subscription * ⓘ

Resource group * ⓘ [Create new](#)

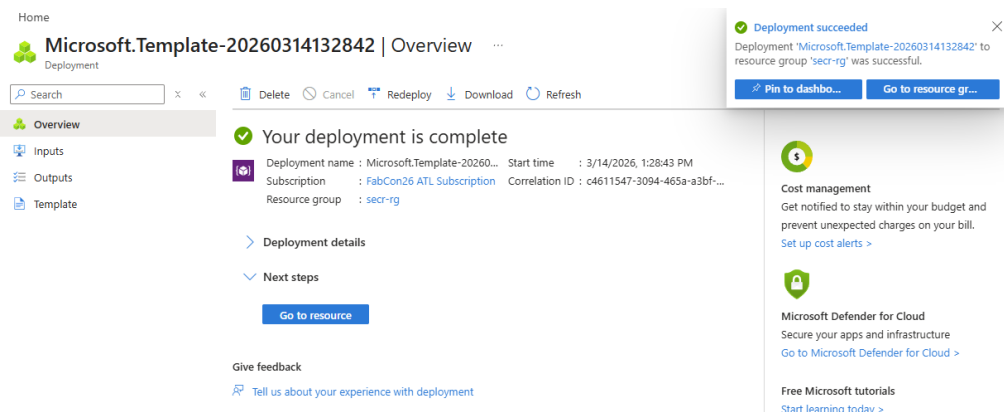
Instance details

Region * ⓘ 

20. Click Review + Create at the button.



21. Once deployment is complete, you should see notification in the top-right corner.

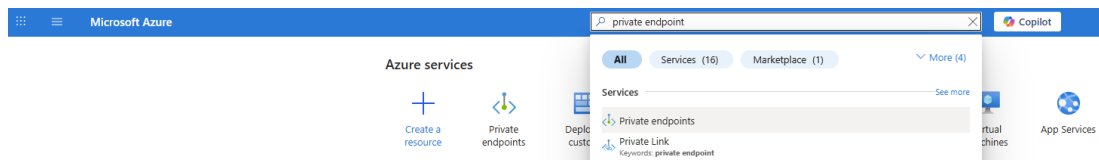


Checkpoint – You have completed setup for private link service for Fabric workspace.

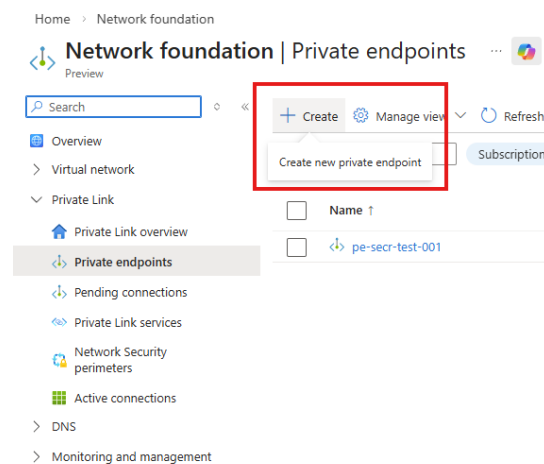
Step 1.3: Create a Private Endpoint for your workspace

The following procedure creates a private endpoint in the virtual network created for you. As part of the process, you associate the private endpoint with the private link for Fabric workspace.

22. In the Search bar at the top, search for private endpoint and click on private endpoint from the result



23. Select **+ Create** in **Private endpoints**



24. On the **Basics** tab of **Create a private endpoint**, enter or select the following information:

- **Subscription:** FabCon26 ATL Subscription
- **Resource group:** Select your resource group assigned to you: **secr-rg**
- **Name:** **pe-ws-<username>-wspl** (example: **pe-ws-secr-user-150-wspl**)
- **Network interface name:** **pe-<username>-wspl-nic** (example: **pe-ws-secr-user-150-wspl-nic**)
- **Region:** West US (do not change)

Home > Network foundation | Private endpoints

Create a private endpoint ...

1 Basics 2 Resource 3 Virtual Network 4 DNS 5 Tags 6 Review + create

Use private endpoints to privately connect to a service or resource. Your private endpoint must be in the same region as your virtual network, but can be in a different region from the private link resource that you are connecting to. [Learn more](#)

Project details

Subscription *	FabCon26 ATL Subscription
Resource group *	secr-rg

[Create new](#)

Instance details

Name *	pe-ws-secr-user-150-wspl
Network Interface Name *	pe-ws-secr-user-150-wspl-nic
Region *	West US

In this step, you will link the Fabric workspace private link you created in step-2 with the private endpoint. Enter the following details and then select **Next**.

- **Connection Method:** Connect to an Azure resource in my directory.
- **Subscription:** FabCon26 ATL Subscription
- **Resource type:**
Search for Fabric and select
Microsoft.Fabric/privateLinkServicesForFabric

✓ Basics 2 Resource 3 Virtual Network 4 DNS 5 Tags 6 Review + create

Private Link offers options to create private endpoints for different Azure resources, like your private link service, a SQL server, or an Azure storage account. Select which resource you would like to connect to using this private endpoint. [Learn more](#)

Connection method ☒ Connect to an Azure resource in my directory. ☐ Connect to an Azure resource by resource ID or alias.

Subscription * FabCon26 ATL Subscription

Resource type * Select resource type

fabrik ←

Microsoft.Fabric/privateLinkServicesForFabric

- **Resource:** Choose the Fabric workspace private link (WS PL) resource you created in **step-1.2**. Please ensure that you select the PL **you** provisioned. The list will show all private links including those provisioned by other participants. Please select the one which you provisioned.
It should be in the format: pls-ws-<username>-wspl (example: pls-ws-secr-user-150-wspl)
- **Target sub-resource:** Select **Workspace**

Home > Network foundation | Private endpoints

Create a private endpoint ...

✓ Basics 2 Resource 3 Virtual Network 4 DNS 5 Tags 6 Review + create

Private Link offers options to create private endpoints for different Azure resources, like your private link service, a SQL server, or an Azure storage account. Select which resource you would like to connect to using this private endpoint. [Learn more](#)

Connection method ☒ Connect to an Azure resource in my directory. ☐ Connect to an Azure resource by resource ID or alias.

Subscription * FabCon26 ATL Subscription

Resource type * Microsoft.Fabric/privateLinkServicesForFabric

Resource * pls-ws-secr-user-150-wspl

Target sub-resource * workspace

25. Virtual Network: vnet-fabcon-sec

Subnet: Expand the drop down and select **your** subnet in the format: **snet-username** (example: snet-secr-user-150)

You can use the search filter to narrow it down like this

✓ Basics ✓ Resource **3 Virtual Network** 4 DNS 5 Tags 6 Review + create

Networking

To deploy the private endpoint, select a virtual network subnet. [Learn more](#)

Virtual network

Subnet *

Network policy for private endpoints

Compatible subnets

snet-secr-user-150 (10.0.7.32/27)
27 available IP addresses

⚠ Network policy changes require Microsoft.Network/virtualNetworks/subnets/write permission

Private IP configuration

☒ Dynamically allocate IP address
☐ Statically allocate IP address

Application security group

Configure network security as a natural extension of an application's structure. ASG allows you to group virtual machines and define network security policies based on those groups. You can specify an application security group as the source or destination in an NSG security rule. [Learn more](#)

+ Create

Application security group

There are other configuration items that you can leave as-is:

- Network policy for private endpoints: Disabled (leave it as is)
- **Private IP configuration:** Dynamically allocate IP address (**Do Not Change**)
- **Application security group:** Leave it blank

Home > Network foundation | Private endpoints

Create a private endpoint ...

✓ Basics ✓ Resource **3 Virtual Network** 4 DNS 5 Tags 6 Review + create

Networking

To deploy the private endpoint, select a virtual network subnet. [Learn more](#)

Virtual network

Subnet *

Network policy for private endpoints Disabled [\(edit\)](#)

⚠ Network policy changes require Microsoft.Network/virtualNetworks/subnets/write permission

Private IP configuration

☒ Dynamically allocate IP address
☐ Statically allocate IP address

Application security group

Configure network security as a natural extension of an application's structure. ASG allows you to group virtual machines and define network security policies based on those groups. You can specify an application security group as the source or destination in an NSG security rule. [Learn more](#)

+ Create

Application security group

26. In the DNS setting, under **Private DNS Integration**, enter or select the following information (if its pre-selected, make sure the below config is correct and do not change)

- **Integrate with private DNS zone: Yes (Do not change)**
- **Subscription: FabCon26 ATL Subscription**
- **Resource Group: secr-rg**
- **Private DNS zone: *privatelink.fabric.microsoft.com***

✓ Basics ✓ Resource ✓ Virtual Network **4 DNS** 5 Tags 6 Review + create

Private DNS integration

To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint with a private DNS zone. You can also utilize your own DNS servers or create DNS records using the host files on your virtual machines. [Learn more](#)

Integrate with private DNS zone ☒ Yes ☐ No

Configuration name	Subscription	Resource group	Private DNS zone
privatelink-fabric-micros...	FabCon26 ATL Subsc...	secr-rg	privatelink.fabric.microso...

i Existing Private DNS Zones tied to a single service should not be associated with two different Private Endpoints as it will not be possible to properly resolve two different A-Records that point to the same service. However, Private DNS Zones tied to multiple services would not face this resolution constraint.

27. Select **Next: Tags (leave it empty)** then **Next: Review + create**

Home > Network foundation | Private endpoints

Create a private endpoint ...

✓ Basics ✓ Resource ✓ Virtual Network ✓ DNS **5 Tags** 6 Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
	:	2 selected

28. Review all the details that you selected matches your **username**

Home > Network foundation | Private endpoints

Create a private endpoint

✓ Validation passed

✓ Basics ✓ Resource ✓ Virtual Network ✓ DNS ✓ Tags **Review + create**

Basics

Subscription	FabCon26 ATL Subscription
Resource group	secr-rg
Region	West US
Name	pe-ws-secr-user-150-wspl
Network interface Name	pe-ws-secr-user-150-wspl-nic

Resource

Subscription ID	1f0636c2-dd23-47c6-be1f-283ac038f057 (FabCon26 ATL Subscription)
Link type	Microsoft.Fabric/privateLinkServicesForFabric
Resource group	secr-rg
Resource	pls-ws-secr-user-150-wspl
Target sub-resource	workspace

Virtual Network

Virtual network resource group	secr-rg
Virtual network	vnet-fabcon-sec
Subnet	sn1et-secr-user-150 (10.0.7.32/27)
Network Policies	Disabled
Application security groups	None

DNS

Integrate with private DNS zone?	Yes
Statically allocate Private IP	No
Private DNS zone resource group	secr-rg
Private DNS zone	privatelink.fabric.microsoft.com

Tags

None

You should see the deployment complete successfully:

Home

Microsoft.PrivateEndpoint-20260314153453 | Overview

Deployment

Search x « Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name : Microsoft.PrivateEndpoint-20260314153453 Start time : 3/14/2026, 3:40:09 PM
Subscription : FabCon26 ATL Subscription Correlation ID : d789bfb3-765a-4909-914e-...
Resource group : secr-rg

Deployment details

Resource	Type	Status
✓ DnsZoneGroup-20260314154007	Deployment	OK
✓ pe-ws-secr-user-150-wspl	Private endpoint	OK

Next steps

[Go to resource](#)

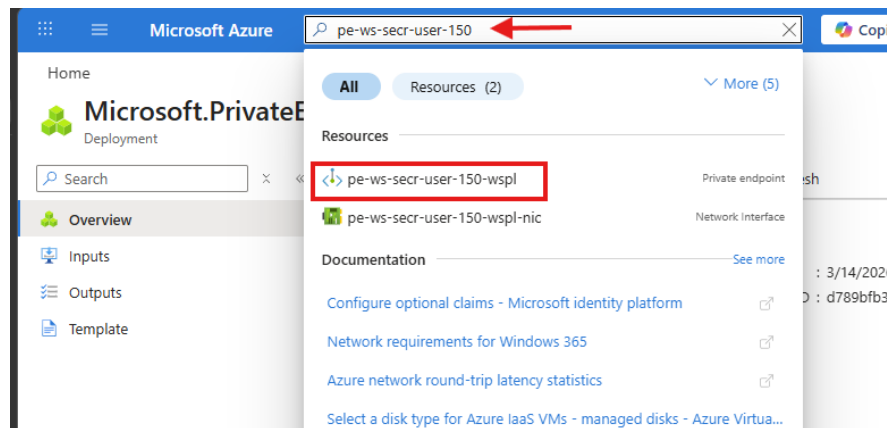
Checkpoint – At this point, you have completed provisioning process for Fabric workspace PL; created a Fabric private endpoint; and associated it with the workspace PL.

Step 1.4: Validate your deployment

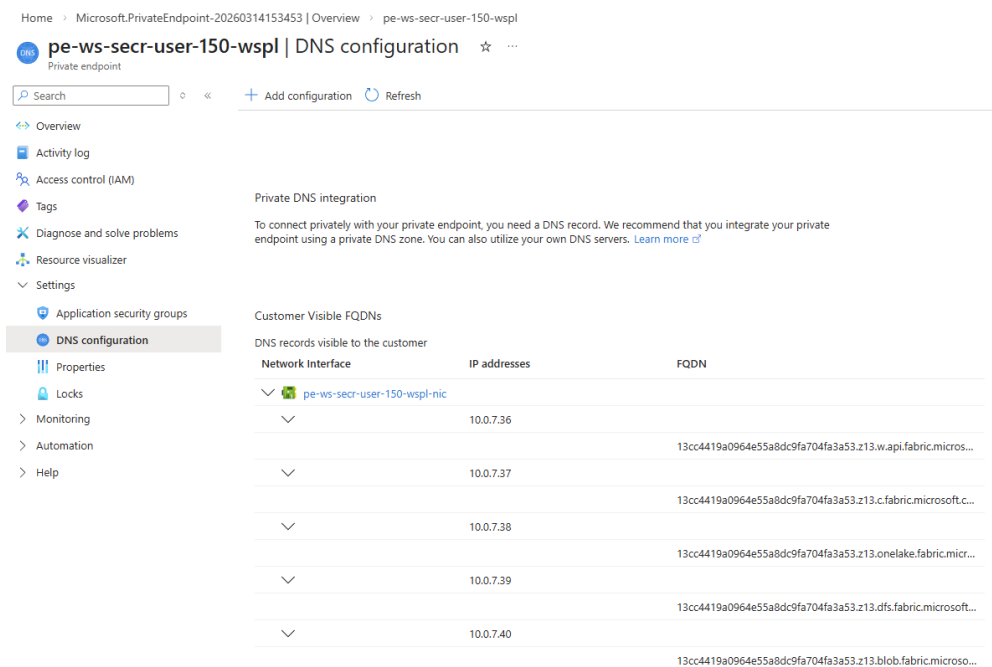
The following steps are optional and do not involve any configuration changes. This is supplemental guidance to verify private endpoints are provisioned correctly.

29. In the Azure portal, navigate to Fabric private endpoint provisioned for your workspace:

Use the search bar at top to locate your resource: **pe-ws-username-wspl**



30. Select **DNS configuration** under **Settings** in the left Nav to list all the private IP addresses associated with workspace-scoped FQDNs.



If you block public access to your workspace, you can now use these workspace FQDNs and IP addresses via a jump-host to access your workspace.

31. If you don't have the Fabric window opened already, in another browser tab, open <https://app.fabric.microsoft.com/> (don't close the Azure tab just yet):
32. Your private link workspace: **ws_username_wspl** (example: **ws_secr-user-150_wspl**). Make sure that the workspace ID you see on the address bar matches with the workspace FQDNS on the Azure DNS config:

The screenshot shows the Microsoft Fabric application interface. The browser address bar displays the URL <https://app.fabric.microsoft.com/group/13cc4419-a096-4e55-a8dc-9fa704fa3a53/?st?experience=fabric-developer>. The workspace name is **ws_secr-user-150_wspl**. Below the workspace name, there is a table titled "Customer Visible FQDNs" showing DNS records visible to the customer.

Network Interface	IP addresses	FQDN
pe-pe-secr-user-150-wspl-nic	10.0.7.36	13cc4419a0964e55a8dc9fa704fa3a53.z13.w.api.fabric.micros...
	10.0.7.37	13cc4419a0964e55a8dc9fa704fa3a53.z13.c.fabric.microsoft.c...
	10.0.7.38	13cc4419a0964e55a8dc9fa704fa3a53.z13.onelake.fabric.micr...
	10.0.7.39	13cc4419a0964e55a8dc9fa704fa3a53.z13.dfs.fabric.microsoft...
	10.0.7.40	13cc4419a0964e55a8dc9fa704fa3a53.z13.blob.fabric.microso...

Summary

This concludes module 1. In this module, you examined how Microsoft Fabric enforces inbound access controls at the workspace level. You validated default inbound networking behaviour, deployed a workspace-scoped Private Link Service using Azure Resource Manager, and associated it with a private endpoint and private DNS integration. You also reviewed how private workspace can be accessed from public workspace using managed private endpoint.

This concludes Module 1 – Inbound Protection.

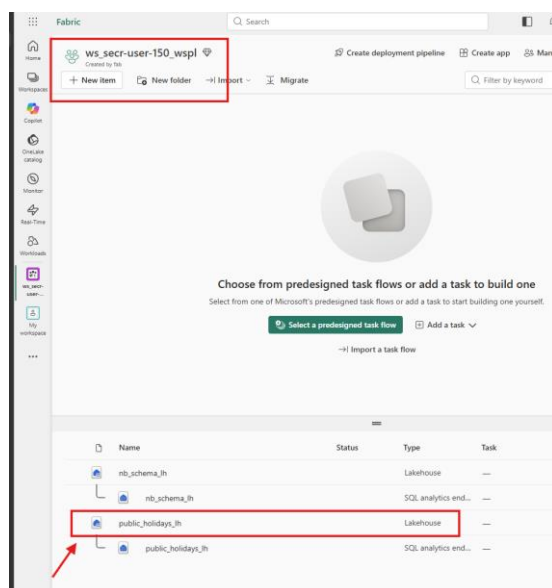
Optional Exercises to test Workspace Private Links

Goal of these exercises is to demonstrate how workspace private links behave once public access inbound access is blocked.

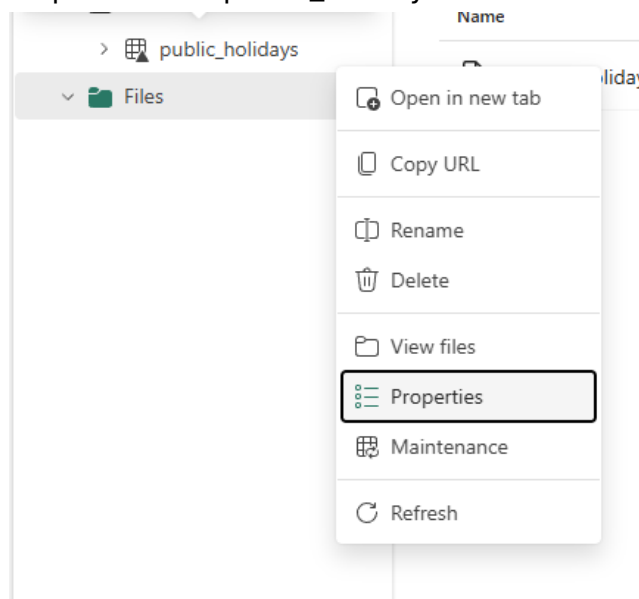
Exercise 1 – Block public internet access

In the configuration we enabled earlier, you have provisioned workspace private link for a workspace, but public endpoint access is not blocked. In this exercise, you will enable hard block on public access. Once this is in place, there are implications for workspace which must be considered.

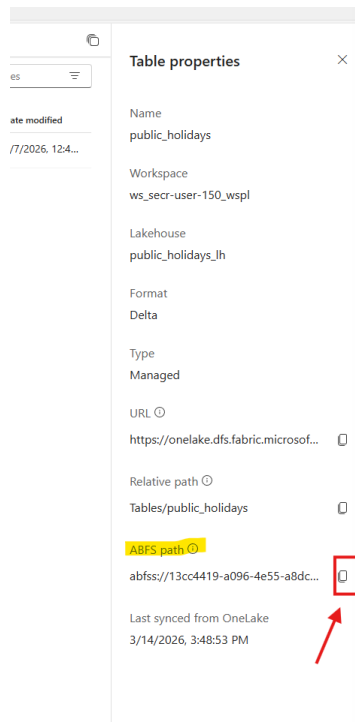
33. Go back to Fabric workspace and open the **public_holidays_lh** lakehouse in your **_wspl** workspace



34. Click on the three elipses next to public_holidays table and select **Properties**



35. In the Table properties, copy the ABFSS path



Example: abfss://13cc4419-a096-4e55-a8dc-9fa704fa3a53@onelake.dfs.fabric.microsoft.com/13ee398c-dc62-4370-9236-9542886afc75/Tables/public_holidays

This ABFSS path is in the format:

`abfss://{WorkspaceID}@onelake.dfs.fabric.microsoft.com/{LakehouseID}/Tables/public_holidays`

`13cc4419-a096-4e55-a8dc-9fa704fa3a53` is the **WorkspaceID** in our example.

`13ee398c-dc62-4370-9236-9542886afc75` is the **LakehouseID** in our example.

You must construct the **WorkspaceFQDN** for OneLake in the following format:

WorkspaceFQDN = `{workspaceIDwithouthyphen}.z{xy}.onelake.fabric.microsoft.com`

Where **xy** is the **first two characters** in your **WorkspaceID**.

For our example, **WorkspaceFQDN** will be (you must construct yours accordingly – **note there is no hyphen in the workspace ID**):

`13cc4419a0964e55a8dc9fa704fa3a53.z13.onelake.fabric.microsoft.com`

Now add this **WorkspaceFQDN** to your ABFSS path in this format:

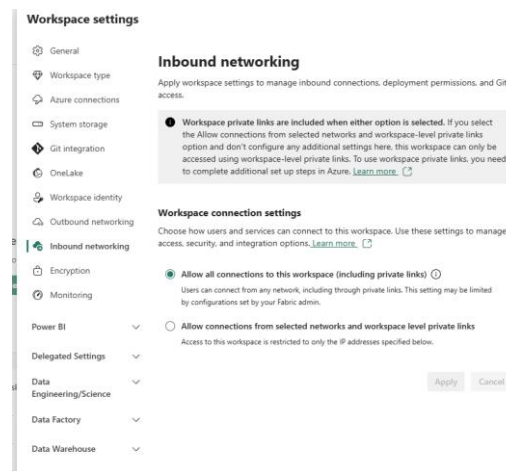
`abfss://{WorkspaceID}@{WorkspaceFQDN}/{LakehouseID}/Tables/public_holidays`

In our example, ABFSS path will be like:

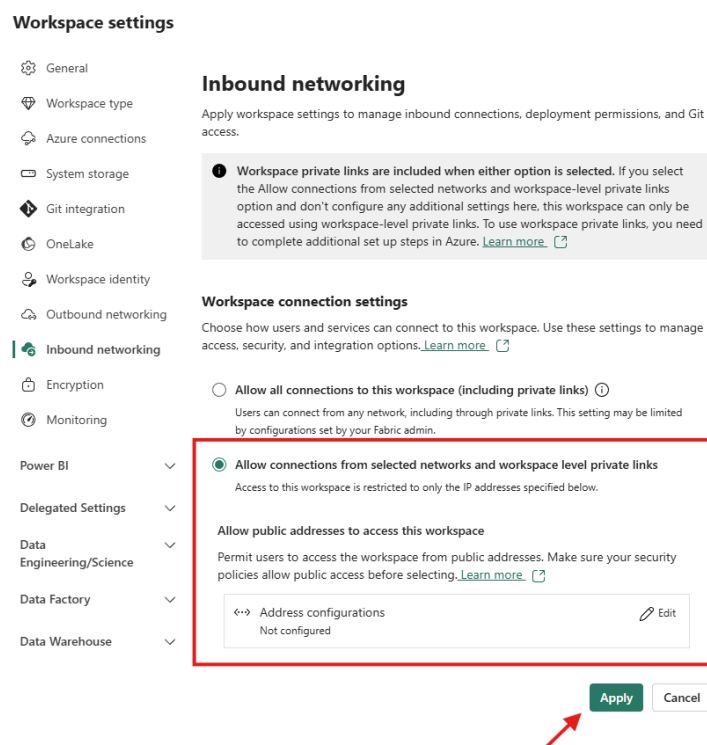
abfss://13cc4419-a096-4e55-a8dc-9fa704fa3a53@13cc4419a0964e55a8dc9fa704fa3a53.z13.onelake.fabric.microsoft.com/13ee398c-dc62-4370-9236-9542886afc75/Tables/public_holidays

Make a note of this URL - You will need this ABFSS path later for our exercise.

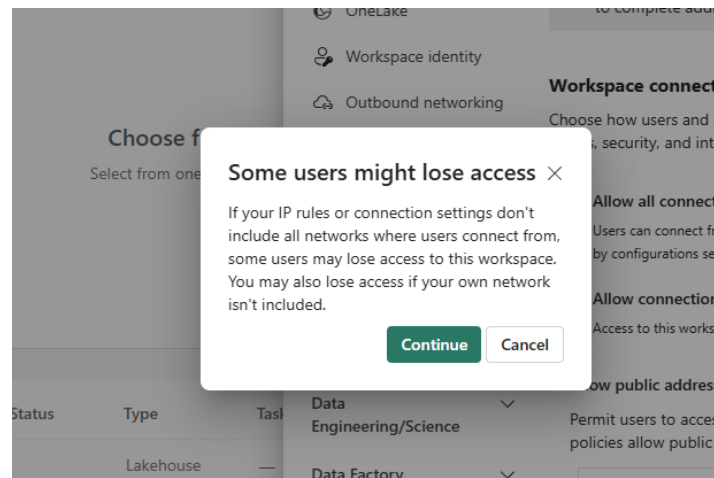
36. Now that you have created private endpoints and noted your ABFSS path, we can test it by blocking the public endpoints with the following step -
- In Fabric, go to **Workspace Settings > Inbound Networking** in the Fabric workspace ending in _wspl.



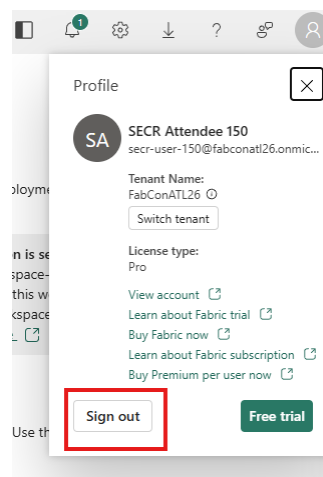
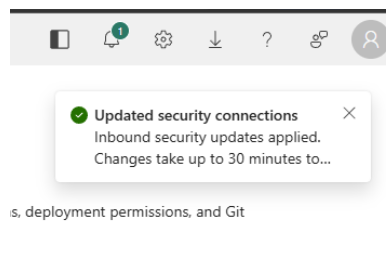
37. Toggle the switch from “Allow all connections” to “**Allow connections from selected networks and workspace level private links**” and click “**Apply**”.



You should see a **warning** pop up like this. Acknowledge that message and click **“Continue”**



38. You will see a notification on top right corner that inbound settings are being applied and may take 30 minutes.



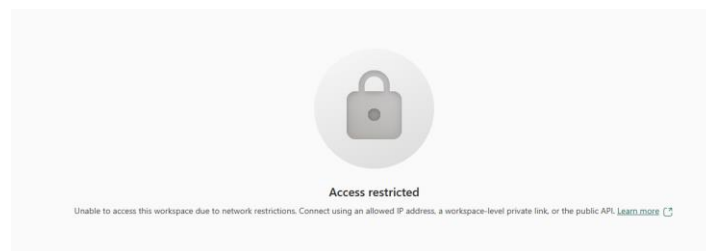
Exercise 2: Accessing a workspace which has private link enabled and public internet access is blocked

In this exercise, you are going to learn how to connect to your private workspace in Fabric from other public workspaces and access the data using Notebooks.

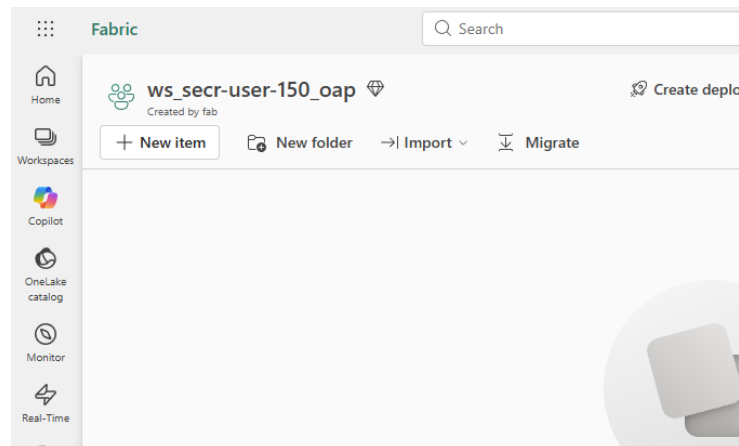
For this pattern to work, you are going to create a “**Managed Private Endpoint**” from public workspace (**OAP workspace**) to the private workspace. You will also log into Azure Portal again to approve those managed private endpoints to authorize this connectivity from the public workspace. Once that is established, you will access the data from a private workspace using a notebook in public workspace (**OAP workspace**).

Step 2.1: Create Managed Private Endpoint from Public Workspace to Private Workspace

1. Open a new InPrivate or InCognito window and login to Fabric:
<https://app.fabric.microsoft.com> with your credentials
2. Open your private workspace: **ws_username_wspl** (example: **ws_secr-user-150_wspl**)
3. You should get an error like this below telling you that you cannot access this workspace from your network - “Access restricted”. (Note: *if you don’t see the error but you are able to see the content of the workspace, check the Workspace Settings and Inbound network settings to make sure that you are allowing connections only from selected networks and private links. You can continue this exercise and check back later and refresh your browser window, you will get an error like this*).



4. Now open your OAP workspace: **ws_username_oap** (Example: **ws_secr-user-150_oap**)



5. Go to **Workspace Settings > Outbound Networking**

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption
- Monitoring

Power BI

Delegated Settings

Data Engineering/Science

Data Factory

Data Warehouse

Network security

Managed private endpoints let people securely connect to an Azure resource or Private Link service. [Learn more](#)

Managed private endpoints

Managed private endpoints are currently available for notebooks and Spark job definitions. [Learn more](#)

+ Create Refresh Delete



No endpoints to show

Outbound access protection

Block outbound public access ☐ Off

Block all outbound connections from workspace and only allow connections through private end points and connection rules. This feature is only available for limited items. Creation of other items will be disabled once this feature is turned on. Please wait for 15 mins for the setting change to take effect. [Learn more](#)

Allow Git Integration ☐ Off

Use all Git features, like commit and sync, for the repository connected to this workspace. If this is off, Git integration will be blocked.

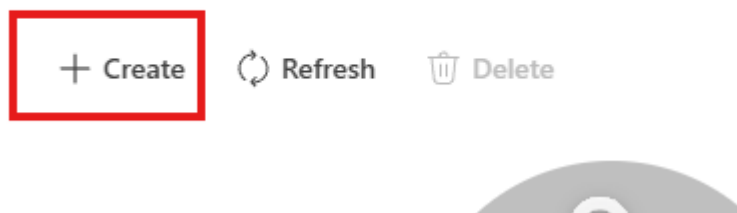
6. Click “Create” under Managed Private Endpoints:

Network security

Managed private endpoints let people securely connect to a service. [Learn more](#) 

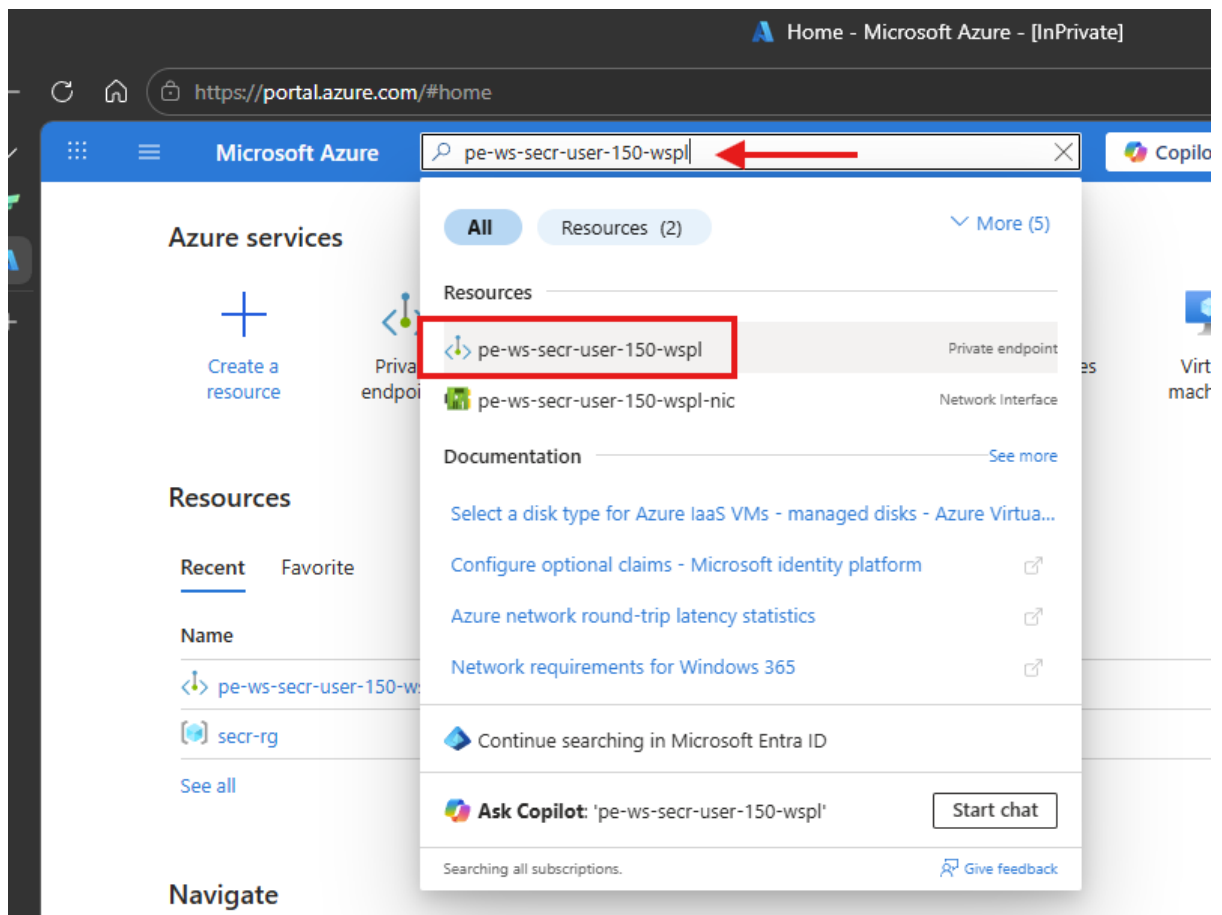
Managed private endpoints

Managed private endpoints are currently available for notebooks. [Learn more](#) 

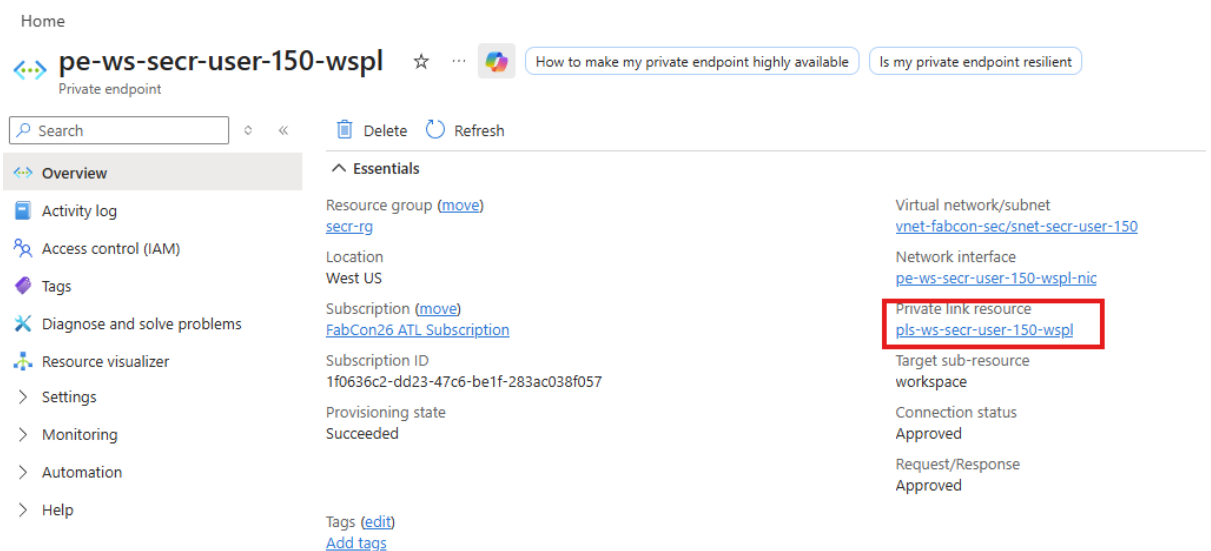


7. Enter Managed private endpoint name as: **mpe-ws-`<username>`-wspl**(example: **mpe-ws-secr-user-150-wspl**)
8. For Resource Identifier, you have to login back to Azure Portal in a new InPrivate window and open the private endpoint that you created earlier, get the Resource ID of your workspace private link service and paste that in Fabric (so, don't close your Fabric window yet):

Go to Azure Portal and you can use the search bar at the top to easily locate your private endpoint resource: **pe-ws-username-wspl**



9. Open the private endpoint and click on the “Private Link Resource” hyperlink.



10. From this page, you can copy the Resource ID of your workspace private link service:

Home > pe-ws-secr-user-150-wspl

pls-ws-secr-user-150-wspl ☆ ... Show me the settings for this Private Link Service. How do I troubleshoot connectivity issues with this resource? +1

Search Refresh Delete Open in mobile

Overview

- Activity log
- Access control (IAM)
- Resource visualizer
- Settings
- Automation
- Help

Essentials

Resource group (move) [secur-rg](#)

Location
Global

Subscription (move) [FabCon26 ATL Subscription](#)

Subscription ID
1f0636c2-dd23-47c6-be1f-283ac038f057

Tags (edit) [Add tags](#)

Resource id
[/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGrou...](#) Copy to clipboard

Type
Microsoft.Fabric/privateLinkServicesForFabric

Microsoft.Fabric private link services for fabric

System data ---

Properties

Tenant Id [f527843b-95a5-498e-b6d4-e8895b9816c6](#)

Workspace object Id ---

Private endpoint con... [View value as JSON](#)

Provisioning state [Succeeded](#)

11. Now go back to Fabric window and paste the Resource ID in the Managed Private Endpoint window:

Create managed private endpoint

Create a managed private endpoint to securely connect this Fabric workspace to your datasources.

Managed private endpoint name ⓘ

mpe-ws-secr-user-150-wspl

Resource identifier ⓘ

'providers/Microsoft.Fabric/privateLinkServicesForFabric/pls-ws-secr-user-150-wspl'

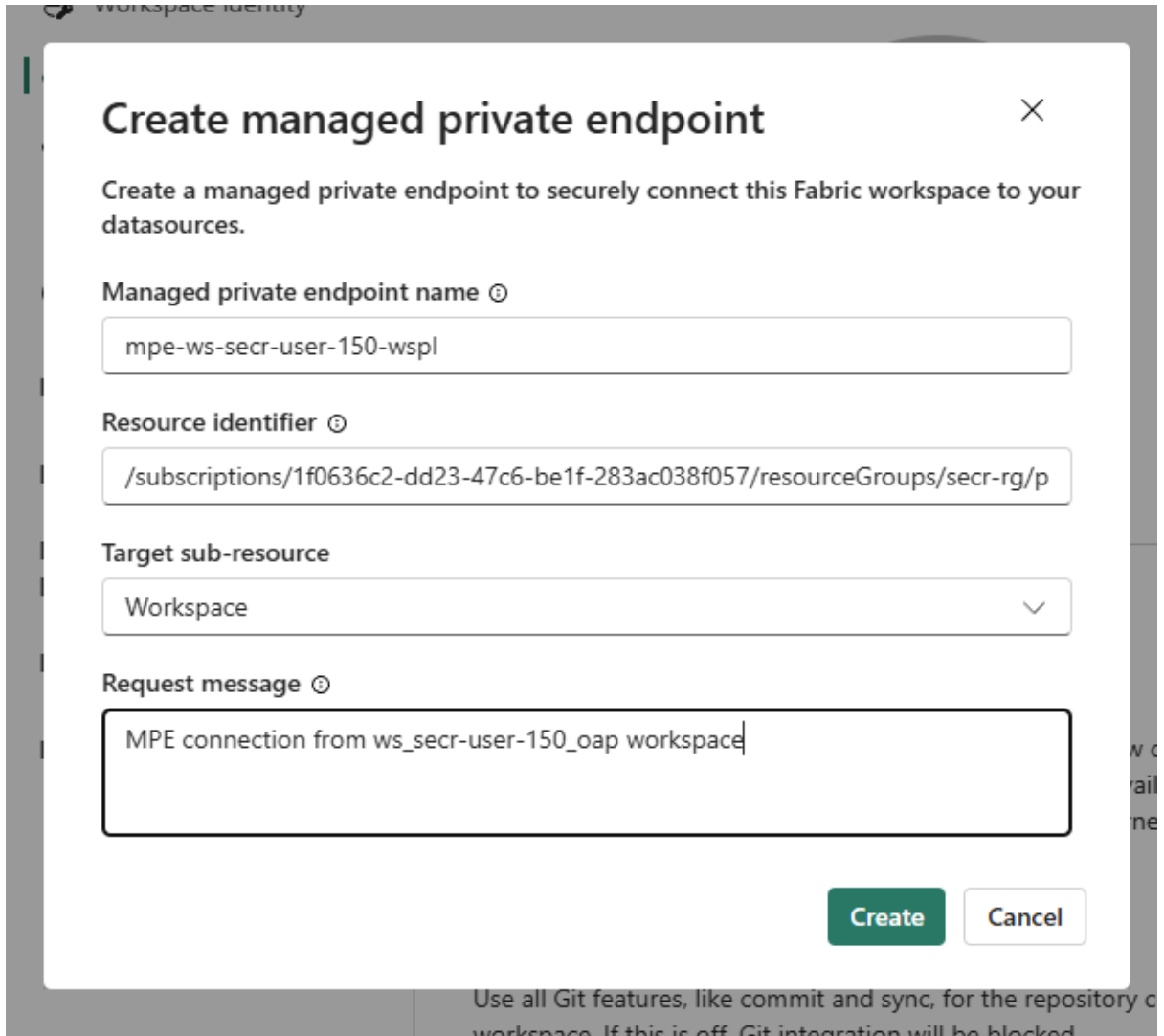
Target sub-resource

Request message ⓘ

Create Cancel

Use all Git features, like commit and sync, for the repository connection to this workspace. If this is off, Git integration will be blocked.

12. Select the Target Sub-Resource as **Workspace**
13. In the Request message, enter some meaningful message that can identify your public workspace like “MPE connection from ws_username_oap workspace” and click Create



The screenshot shows a modal dialog titled "Create managed private endpoint" with a close button (X) in the top right corner. Below the title is a descriptive sentence: "Create a managed private endpoint to securely connect this Fabric workspace to your datasources." The dialog contains four input fields: "Managed private endpoint name" with the value "mpe-ws-secr-user-150-wspl", "Resource identifier" with the value "/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/p", "Target sub-resource" with a dropdown menu showing "Workspace", and "Request message" with the value "MPE connection from ws_secr-user-150_oap workspace". At the bottom right, there are two buttons: "Create" (green) and "Cancel" (white with a grey border). Below the dialog, a partial line of text is visible: "Use all Git features, like commit and sync, for the repository c workspace. If this is off, Git integration will be blocked."

Create managed private endpoint ✕

Create a managed private endpoint to securely connect this Fabric workspace to your datasources.

Managed private endpoint name ⓘ

mpe-ws-secr-user-150-wspl

Resource identifier ⓘ

/subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/p

Target sub-resource

Workspace ▾

Request message ⓘ

MPE connection from ws_secr-user-150_oap workspace

Create Cancel

Use all Git features, like commit and sync, for the repository c workspace. If this is off, Git integration will be blocked.

14. You will see a message called “Provisioning” in your screen like this (it may take about 5 minutes to provision and change the status to Approve – you can click Refresh to check after 5 minutes):

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption
- Monitoring
- Power BI
- Delegated Settings
- Data Engineering/Science
- Data Factory
- Data Warehouse

Network security

Managed private endpoints let people securely connect to an Azure resource or Private Link service. [Learn more](#)

Managed private endpoints

Managed private endpoints are currently available for notebooks and Spark job definitions. [Learn more](#)

+ Create **Refresh** Delete

☐	Name ↓	Activation	Approval
☐	mpe-ws-secr-user-150-wspl	Provisioning	-

Outbound access protection

Block outbound public access ☐ Off

Block all outbound connections from workspace and only allow connections through private end points and connection rules. This feature is only available for limited items. Creation of other items will be disabled once this feature is turned on. Please wait for 15 mins for the setting change to take effect. [Learn more](#)

Allow Git Integration ☐ Off

Use all Git features, like commit and sync, for the repository connected to this workspace. If this is off, Git integration will be blocked.

You may take a small break for 5 minutes at this stage

15. After about 5 minutes break, you should see the Activation “Succeeded” and Approval “Pending”.

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption
- Monitoring
- Power BI
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- Data Warehouse

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Managed private endpoints

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+ Create Refresh Delete

	Name ↓	Activation	Approval
	mpe-ws-secr-user-149-wspl	Succeeded	Pending

Outbound access protection

Block outbound public access Off

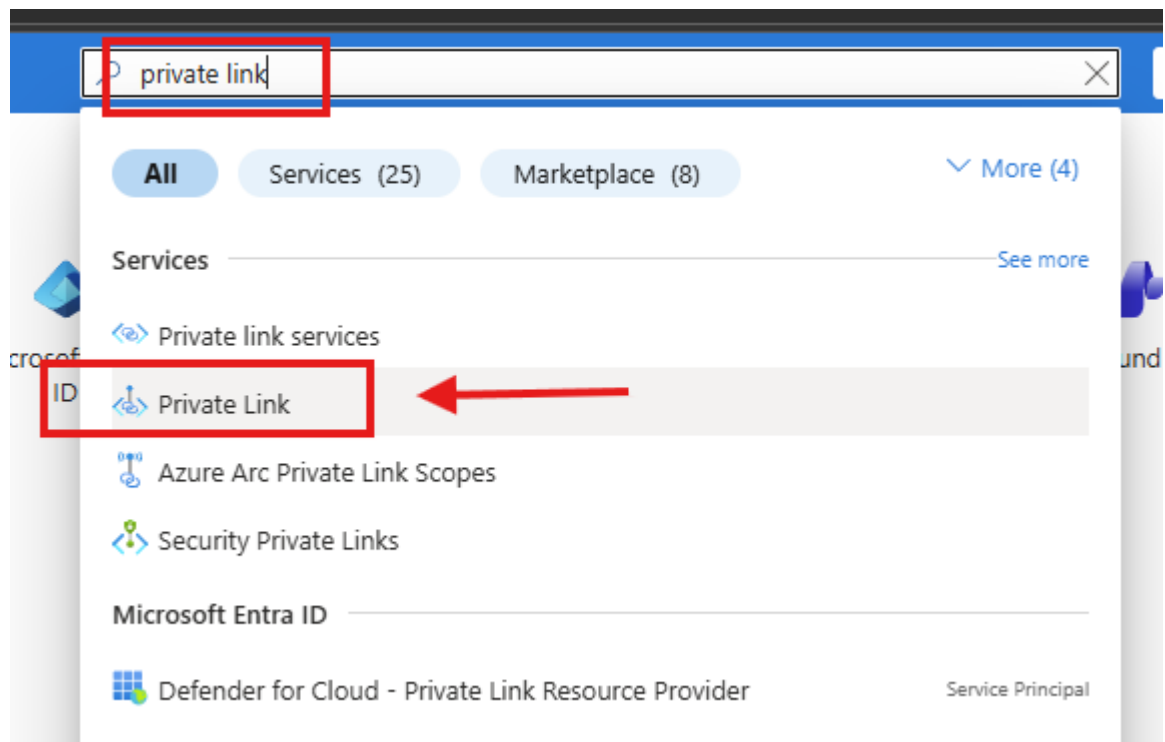
Block all outbound connections from workspace and only allow connections through private end points and connection rules. This feature is only available for limited items. Creation of other items will be disabled once this feature is turned on. Please wait for 15 mins for the setting change to take effect. [Learn more](#)

Allow Git Integration Off

Use all Git features, like commit and sync, for the repository connected to this workspace. If this is off, Git integration will be blocked.

16. You must go back to Azure Portal to approve this managed private endpoint connection. Login to Azure Portal (if not already open)

17. In Azure, search for “Private Link” and Click on the second link in the result that says “Private Link” (see screenshot below)



This will open the Private Link Centre in Network Foundation.

Home > Network foundation

Network foundation | Private Link overview

Preview

search

Overview

> Virtual network

> Private Link

Private Link overview

> Private endpoints

> Pending connections

> Private Link services

> Network Security perimeters

> Active connections

> DNS

> Monitoring and management

Private Endpoint

Network foundation

Connect to a Private Link resource and bring that resource directly into your virtual network.

✓ Access services privately within your virtual network.

✓ Enhance security and compliance.

✓ Prevent data exfiltration over the internet.

[Learn more](#)

Create

View resources

Provide secure access to your resources

Private Link Service

Network foundation

Consume private, secure and scalable services from your virtual network. [Learn more](#)

Create

View resources

Network security for cloud services

Network security perimeters

Network foundation

Restrict public exposure of Azure Services and manage inbound and outbound access for resources within perimeter. [Learn more](#)

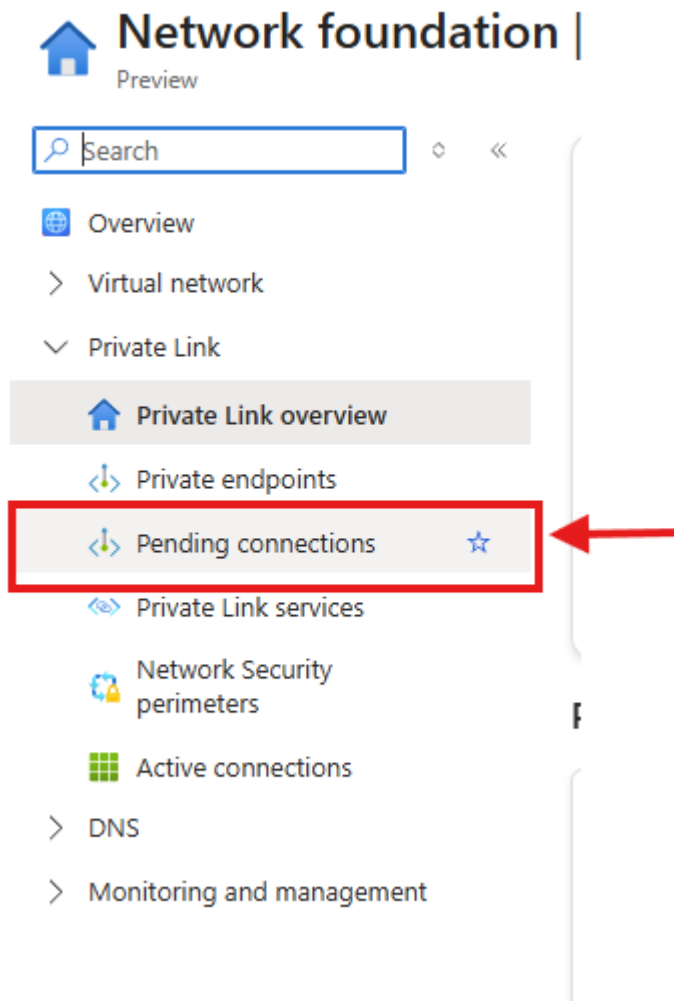
Create

View resources

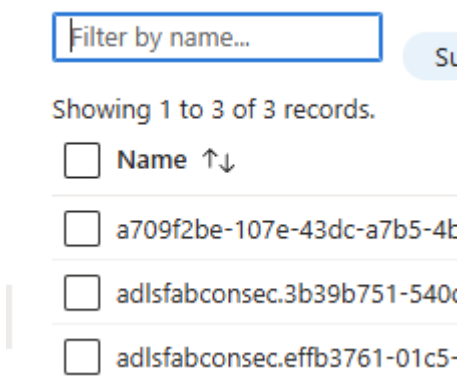
Give feedback

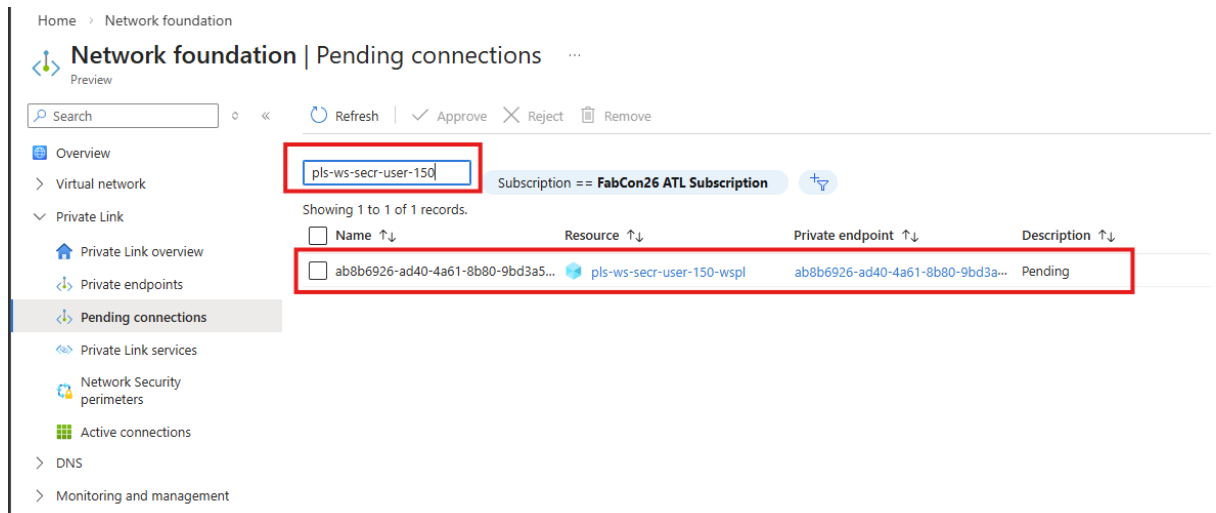
[Share feedback about your experience](#)

18. Click on “Pending Connections” in the left nav:



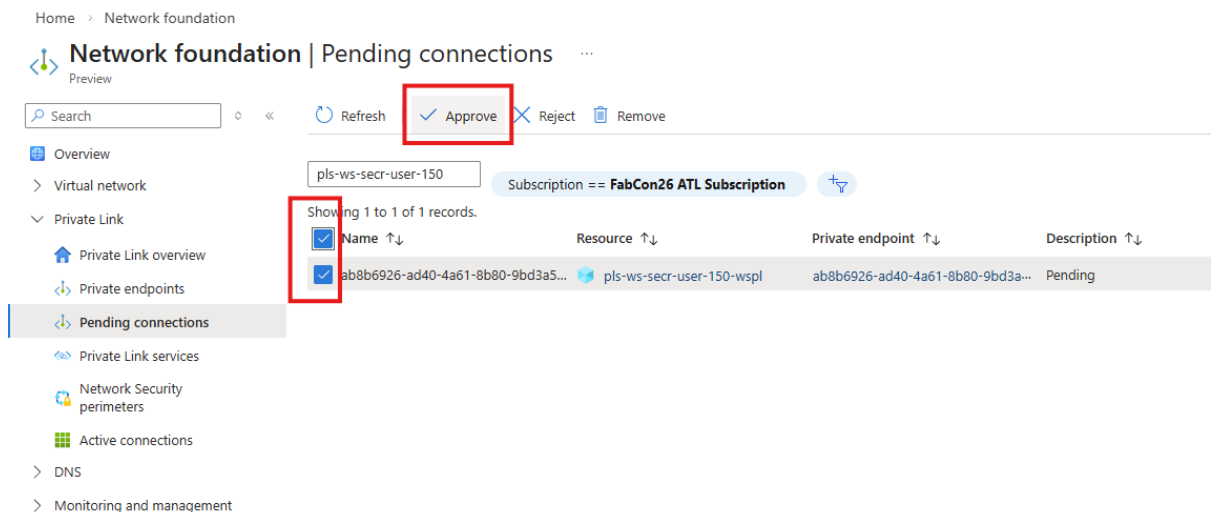
19. There will be a lot of connections pending from all the attendees. **You should search for your connection only and approve.** In the “Filter by Name” text box, type the name of your **workspace private link service: pls-ws-username-wspl** (example: pls-ws-secr-user-150-wspl):





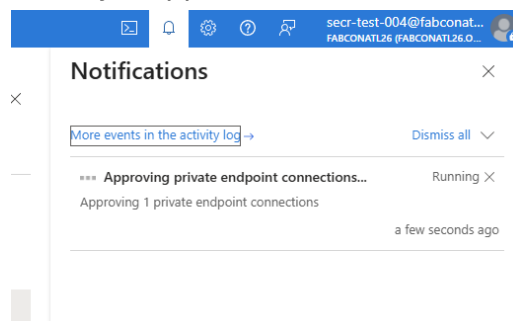
That should bring your managed private endpoint connection request to approve.

20. Do not click on any hyperlink. Just select the checkbox and click “Approve” at the top.



21. You will get a dialog box that asks for confirmation. Please verify your MPE connection name and provide a meaningful message like “Approved by username” and click “Yes”

22. You should see “Approving” message in the Notification bar at the top right corner and after a while it says “Approved”



23. Now go to back to Fabric workspace settings -> Outbound networking and click “Refresh” a few times

Workspace settings

- General
- Workspace type
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Outbound networking**
- Inbound networking
- Encryption
- Monitoring

Network security

Managed private endpoints let people securely connect to an Azure resource or Private Link service. [Learn more](#)

Managed private endpoints

Managed private endpoints are currently available for notebooks and Spark job definitions. [Learn more](#)

+ Create **Refresh** Delete

Name	Activation	Approval
mpe-ws-secr-user-150-wspl	Succeeded	Approved

Outbound access protection

Block outbound public access ☐ Off

Block all outbound connections from workspace and only allow connections through private end points and connection rules. This feature is only available for limited items. Creation of other items will be disabled once this feature is turned on. Please wait for 15 mins for the setting change to take effect. [Learn more](#)

Allow Git Integration ☐ Off

Use all Git features, like commit and sync, for the repository connected to this workspace. If this is off, Git integration will be blocked.

Power BI ☐

Delegated Settings ☐

Data Engineering/Science ☐

Data Factory ☐

Data Warehouse ☐

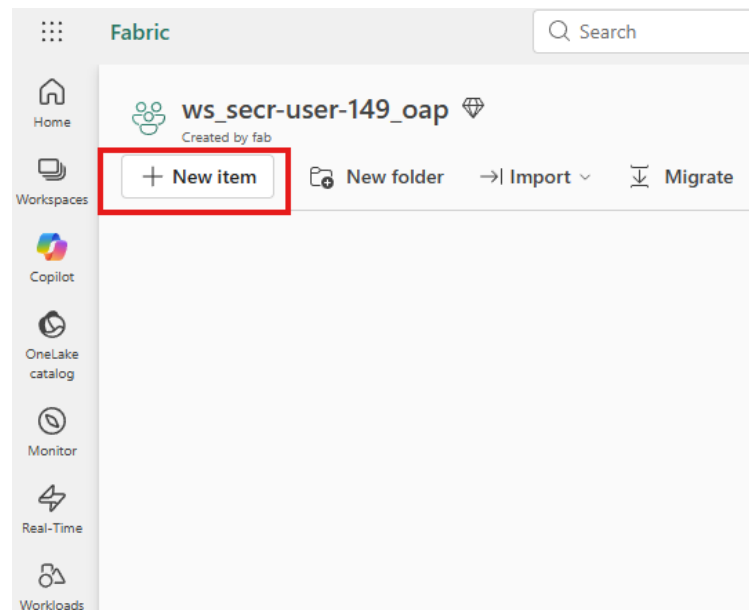
After a few clicks, you should see the Approval Status reflected here as “Succeeded”.

This completes this exercise for enabling cross-workspace access where public endpoint access to a workspace is blocked.

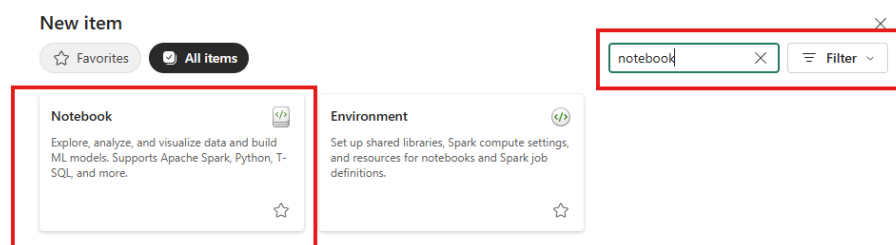
Step 2.2: Access the private Lakehouse from public workspace using a notebook

We have pre-created a Lakehouse in your private workspace. In this step, you will create a new python notebook in the OAP workspace to access the private Lakehouse.

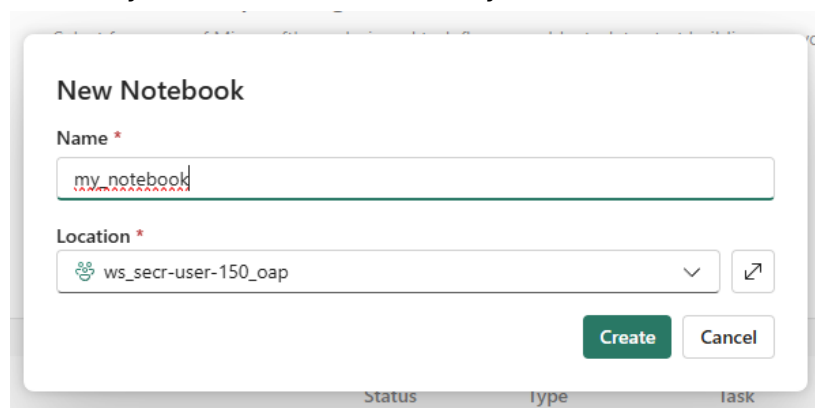
24. Open the OAP workspace and click **New Item**



25. Search for notebook and click on notebook item to create a notebook



26. Provide a name for your notebook such as “my_notebook” and click “Create”



27. Paste the following code into a new cell and run:

```
abfss_path = "replace-with-your-abfss-path"
# Read Delta Lake table from the restricted lakehouse by using the workspace
DNS-based ABFSS URI
df = spark.read.format("delta").load(abfss_path)
df.show(10)
```

In this code snippet, please replace the `abfss_path` with your path to the private workspace that you constructed in [Step 1.4 \(step 36\)](#).

It will be in this format:

```
"abfss://{WorkspaceID}@{WorkspaceFQDN}/{LakehouseID}/Tables/customers"
```

Where

```
WorkspaceID = ws_secr-username_wspl workspace ID
```

```
WorkspaceFQDN = workspaceIDwithouthyphen.z{xy}.onelake.fabric.microsoft.com
```

```
LakehouseID = The Lakehouse ID for public_holidays_lh in ws_secr-username_wspl
workspace
```
